

Xinran ZHAO

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EDUCATION

- **City University of Hong Kong** *Aug. 2020 - Present*
Hong Kong SAR
Ph.D. in Electrical Engineering
 - Supervisor: [Prof. Lin DAI](#)
 - GPA: 4.15/4.30
- **Huazhong University of Science and Technology** *Sep. 2016 - Jun. 2020*
Wuhan, China
Bachelor in Electronics and Information Engineering
 - GPA: 3.86/4.0
 - Thesis: "Random access toward massive machine-to-machine communications: Performance analysis and protocol design", supervised by [Dr. Yuyu GAO](#).

RESEARCH INTERESTS

- **Decentralized multiple access:** Modeling, analysis and optimal design.
- **Next-generation wireless communication networks:** Massive Internet of Things, low-latency communications, and distributed learning-based access design.

ACADEMIC EXPERIENCE

- **Learning-Based Random Access: Design and Optimization** *Jul. 2024 - Present*
 - A simple Multi-Armed-Bandit (MAB) learning framework was presented for performance optimization of random access, based on which two random access schemes were proposed for throughput maximization, that is, MTOA-L with local rewards and MTOA-G with global rewards.
 - The proposed MTOA-L and MTOA-G were shown to both achieve the maximum throughput of 1, though with different short-term fairness performance.
 - It was demonstrated that by leveraging the queueing-theoretical analysis, the tradeoff between throughput and short-term fairness of MTOA-L and MTOA-G can be characterized and optimized by tuning the key parameters.
- **Random Access: Unified Modeling, Optimization and Comparison** *Oct. 2022 - Jun. 2024*
 - A unified analytical framework for random access was established, which incorporates various access design features including sensing-free or sensing-based, connection-free or connection-based, and backoff.
 - The unified framework enables not only the delay characterization and optimization for a given random access scheme, but also a comprehensive comparison of optimal delay performance across different schemes.
 - Useful criteria, such as the upper-bound of sensing time for beneficial sensing, were obtained and further applied to cellular networks.
- **Connection-Based Aloha: Modeling and Performance Optimization** *Aug. 2020 - Sep. 2022*
 - A scalable model was developed for connection-based Aloha.
 - Both the throughput performance and queueing delay performance were characterized, optimized, and compared with those for connection-free Aloha.
 - Conditions for beneficial connection establishment were discussed and further applied to cellular networks.

TEACHING EXPERIENCE

• Tutor for Final Year Projects

- Optimal Backoff Tuning for Massive Access of Machine-type Devices in 5G Networks Fall 2024
- Massive Access for Beyond-5G Communications Networks: Connection-based or Connection-free? Fall 2021, Fall 2022, Spring 2024
- Energy-Efficient Massive Access for Machine-to-Machine Communications Fall 2020

• Teaching Assistant

- Data Engineering and Learning Systems Spring 2024
- Mobile Communication and Networks Spring 2022, Fall 2022, Fall 2023
- Principles of Communications Fall 2021, Spring 2023

PROFESSIONAL EXPERIENCE

• Intelligent Control System of Tobacco Baking Room

Mar. 2019 - Jan. 2020

Team leader of a collaboration project between Dian Group (student tech team in HUST) and China Tobacco Wuhan, China

- **Back-End Development:** Developed the server-side APIs to implement the functions of data reception from IoT terminals, status monitoring of baking rooms, historical data statistics, remote control, etc.
- **System Debugging and Deployment:** Conducted the joint debugging between the server and IoT embedded terminals, the communications between which were built upon the LoRaWAN architecture. Deployed the system in practical tobacco-baking scenarios.
- **Skills:** Python Django, MySQL, Redis and LoRaWAN.

• Online Exam System

Sep. 2018 - Jan. 2020

Member of a project at Dian Group

Wuhan, China

- **Back-End Development:** Developed the server-side APIs to realize the functions of question bank management, examinee information management, online examination, grading, etc.
- **Skills:** Python Flask and MySQL.

PUBLICATIONS

J=JOURNAL, C=CONFERENCE, S=IN SUBMISSION

- [S1] Xinran Zhao and Lin Dai, "Throughput-optimal random access: A queueing-theoretical analysis for learning-based access design," submitted for publication. (Preprint: <https://arxiv.org/abs/2504.03178>)
- [J2] Xinran Zhao and Lin Dai, "To sense or not to sense: A delay perspective," to appear in *IEEE Transactions on Communications*.
- [J1] Xinran Zhao and Lin Dai, "Connection-based Aloha: Modeling, optimization, and effects of connection establishment," *IEEE Transactions on Wireless Communications*, vol. 23, no. 2, pp. 1008–1023, Feb. 2024.
- [C1] Xinran Zhao and Lin Dai, "To sense or not to sense: A delay perspective," *IEEE International Conference on Communications*, Denver, CO, USA, 2024, pp. 2318–2323.

AWARDS

- Research Tuition Scholarship City University of Hong Kong, Sep. 2023
- EE Graduate Research Seminar Award - 3rd Prize Dept. of EE, City University of Hong Kong, May 2022
- Outstanding Undergraduate Thesis Award Huazhong University of Science and Technology, Jun. 2020

SKILLS

- **Mathematical Tools:** Probability theory, queueing theory, stochastic process, reinforcement learning.
- **Programming Languages:** Python, MATLAB, C.
- **Web Technologies:** Python Flask, Python Django, MySQL, Redis.

ADDITIONAL INFORMATION

- **Languages:** Mandarin (native) and English (IELTS 7.0)